Now try plugging in some numbers for α and β. Make sure they are normalized to 1. For example, $\alpha = \frac{1}{\sqrt{2}}$, $\beta = \frac{1}{\sqrt{2}}$.

This simple example is a good way to understand the properties of density matrices. You can refer back to it as we look at the more complex example of an entangled state.

Suppose we know the wave function of a composite system, for example

$$\psi(a, b),$$

but we are only interested in Alice's subsystem. In other words, we want to keep track of everything that Alice can ever measure. Do we have to know the whole wave function? Or is there some way to get rid of Bob's variables? The answer to the latter question is yes; we can capture Alice's complete description in terms of a density matrix ρ.

Let's consider an observable **L** of Alice's system. Like any observable, it can of course be represented as a matrix:

$$L_{a'b',ab} = \langle a'b'|\mathbf{L}|ab\rangle.$$

Remember, for the composite system, the pair *ab* is really a single index labeling a basis vector.

When we say, "**L** is an Alice-observable," what we mean is that **L** does nothing to Bob's half of the state-label. This forces some restrictions on the form of **L**. The idea is to filter out (set equal to zero) any of **L**'s matrix elements that have the effect of changing Bob's half of the state-label. In other words, **L** has the special form

$$L_{a'b',ab} = L_{a'a}\ \delta_{b'b}. \qquad (7.18)$$

This simple-looking equation requires some explanation, and you may want to review the material on tensor products in component form, in the